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June 24, 2024

Theorem 0.1.

$$(B \in A) \to \left(B \subseteq \bigcup A\right)$$

Proof.
Assume $(b \in B)$.
To show: $(b \in \bigcup A)$.

$$(b \in B) \implies (b \text{ set}) \quad (\text{by thm } \mathbf{??})$$
$$(B \in A) \ \ \& \ \ (b \in B) \implies (\exists a \in A : b \in a)$$
$$(b \text{ set}) \ \ \& \ \ (\exists a \in A : b \in a) \implies \left(b \in \bigcup A\right)$$

Done.

Theorem 0.2.

$$(B \in A) \to \left(\bigcap A \subseteq B\right)$$

Proof.
Assume $(B \in A)$, $(x \in \bigcap A)$. To show: $(x \in B)$.

$$(x \in \bigcap A) \implies (x \text{ set}) \ \ \& \ \ (\forall a \in A : x \in a)$$

$$(\forall a \in A : x \in a) \ \ \& \ \ (B \in A) \implies (x \in B)$$

Done.

Theorem 0.3.

$$\left(\bigcap \{w\} = w\right)$$

Proof.
$(\subseteq)$
$(x \in \bigcap \{w\}) \implies (\forall m \in \{w\} : x \in m)$.
$(\forall m \in \{w\} : x \in m) \ \ \& \ \ (w \in \{w\}) \implies (x \in w)$.
$(\supseteq)$
Fix x. Assume $(x \in w)$. To show: $(x \in \bigcap \{w\})$.
$(x \in w) \implies (x \text{ set})$.
Claim: $(\forall m \in \{w\} : x \in m)$.
Fix m. Assume $m \in \{w\}$.
$(m \in \{w\}) \implies (m = w)$.
$(x \in w) \ \ \& \ \ (m = w) \implies (x \in m)$
The claim is proven.

$(x \text{ set}) \ \& \ (\forall m \in \{w\} : x \in m) \implies (x \in \bigcap\{w\})$.
Done.

Theorem 0.4. Intersection of a singleton is in it.

$$(A \text{ singleton}) \to \left(\left(\bigcap A\right) \in A\right)$$

Proof.

$$(A \text{ singleton}) \implies (\exists! w : w \in A)$$
$$(w \in A) \ \& \ (!a : a \in A) \implies \{w\} = A \quad \text{(by the previous theorem)}$$
$$(\{w\} = A) \implies \left(\bigcap\{w\} = \bigcap A\right).$$
$$\left(\bigcap\{w\} = w\right) \ \& \ \left(\bigcap\{w\} = \bigcap A\right) \implies \left(\bigcap A = w\right)$$
$$\left(\bigcap A = w\right) \ \& \ (w \in A) \implies \left(\bigcap A \in A\right)$$

Done.

0.1 Theorems about union and intersection

Theorem 0.5.

$$(A \text{ nonempty}) \to \left(\left(\bigcap A\right) \subseteq \left(\bigcup A\right)\right)$$

Proof.

$$(A \text{ nonempty}) \implies (\exists B : B \in A)$$
$$(B \in A) \implies \left(\bigcap A\right) \subseteq B$$
$$(B \in A) \implies \left(B \subseteq \bigcup A\right)$$
$$\left(\bigcap A\right) \subseteq B \ \& \ \left(B \subseteq \bigcup A\right) \implies \left(\bigcap A \subseteq \bigcup A\right)$$

Done.

Theorem 0.6.

$$(A \text{ set}) \wedge (B \subseteq A) \to (B \in \mathcal{P}A)$$

Proof.

$$(A \text{ set}) \ \& \ (B \subseteq A) \implies (B \text{ set}) \quad \text{(by axm ??)}$$
$$(B \text{ set}) \ \& \ (B \subseteq A) \implies (B \in \mathcal{P}A) \quad \text{(by def ??)}$$

Done.

Theorem 0.7.

$$(A \subseteq \mathcal{P}A) \longleftrightarrow \left(\bigcup A \subseteq A\right)$$

Proof.
($\Leftarrow$)
Assume $(\bigcup A \subseteq A)$. To show:$(A \subseteq \mathcal{P}A)$.
Assume $(x \in A)$. To show: $(x \in \mathcal{P}A)$.
Claim:$(x \subseteq A)$.
Assume $(y \in x)$. To show: $(y \in A)$.

$$(y \in x) \ \ \& \ \ (x \in A) \implies \left(y \in \bigcup A\right)$$

$$\left(y \in \bigcup A\right) \ \ \& \ \ \left(\bigcup A \subseteq A\right) \implies (y \in A)$$

The claim is proven.

$$(x \in A) \implies (x \text{ set}) \quad (\text{by thm } \mathbf{??})$$

$$(x \text{ set}) \ \ \& \ \ (x \subseteq A) \implies (x \in \mathcal{P}A)$$

($\Rightarrow$)
Assume $(A \subseteq \mathcal{P}A)$. To show:$(\bigcup A \subseteq A)$.
Assume $(x \in \bigcup A)$. To show:$(x \in A)$.

$$(A \text{ transitive}) \implies (A \subseteq \mathcal{P}A)$$

$$\left(x \in \bigcup A\right) \implies (\exists y \in A : x \in y)$$

$$(y \in A) \ \ \& \ \ (A \subseteq \mathcal{P}A) \implies (y \in \mathcal{P}A)$$

$$(y \in \mathcal{P}A) \implies (y \subseteq A)$$

$$(x \in y) \ \ \& \ \ (y \subseteq A) \implies (x \in A)$$

Done.

Theorem 0.8.

$$(A \text{ transitive}) \longleftrightarrow \left(\bigcup A \subseteq A\right)$$

Proof.
by **??**
Done.

Theorem 0.9.

$$(A \text{ nonempty}) \wedge (A \text{ transitive}) \longrightarrow \left(\left(\bigcap A\right) \subseteq A\right)$$

Proof.
By theorems **??** and **??**.
Done.